

PROVISIONAL PATENT APPLICATION

Title: Cooperative-Architecture Substrate Pipeline System and Method for Local-First Artificial Intelligence with Verified-Cooperative Knowledge Growth via Multi-Specialist Staggered Swarm, Adversarial Fence, Three-Fates Arbitration, Andon-Cord Self-Policing, and Per-Domain Isolation Mechanisms

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TITLE

Cooperative-Architecture Substrate Pipeline System and Method for Local-First Artificial Intelligence with Verified-Cooperative Knowledge Growth via Multi-Specialist Staggered Swarm, Adversarial Fence, Three-Fates Arbitration, Andon-Cord Self-Policing, and Per-Domain Isolation Mechanisms

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to artificial intelligence systems, and more particularly to cooperative-architecture methods and systems for growing, verifying, and managing local-first knowledge substrates on consumer hardware without dependence on paid cloud API services.

Background and Problem Statement

The cooperative-class AI architecture presented herein addresses a fundamental limitation in current large language model deployment: the absence of verified, locally-grown, domain-specific knowledge substrate that can be empirically measured, grown, and validated on consumer hardware without dependence on paid cloud API services.

Prior approaches to improving LLM accuracy rely on: (1) larger model parameter counts requiring expensive cloud infrastructure; (2) retrieval-augmented generation (RAG) from static knowledge bases that cannot grow or self-correct; (3) fine-tuning that modifies model weights rather than growing external verified knowledge; and (4) multi-model ensemble approaches requiring simultaneous access to multiple large models, incurring prohibitive per-token costs.

None of the foregoing approaches provides: (a) a self-policing quality gate that stops substrate growth on quality failure and restarts with wider knowledge sources; (b) a staggered multi-specialist swarm distributing external API rate-limit risk across nine parallel knowledge domains; (c) an adversarial fence that challenges each candidate knowledge unit before substrate entry; (d) a three-voter concordance arbitration mechanism operating on surviving candidates; (e) a per-domain isolation guarantee preventing cross-domain contamination; or (f) a local-first substrate layout enabling full functionality on consumer hardware with no cloud dependency after initial setup.

Empirical Receipt

The canonical plow pipeline architecture claimed herein has been empirically demonstrated to achieve 97.1% (68/70) on

the MMLU-Pro canonical 70-question benchmark using consumer hardware with no paid API keys. Receipt: 316 verified eplets grown across 14 academic domains, 14/14 domains GREEN, 2 Andon-correct quarantines. This receipt constitutes a reduction to practice for the claims herein.

SUMMARY OF THE INVENTION

The present invention discloses innovations in the following areas, each constituting a novel and non-obvious contribution to the art of cooperative-architecture artificial intelligence:

1. Canonical Plow Pipeline Architecture - a nine-stage sequential knowledge growth pipeline
 2. FireGuard Staggered Swarm Dispatch - a rate-limit-aware multi-specialist parallel query mechanism
 3. Anti-Popularity Miner Filter - a dual-threshold cooperative-class quality gate
 4. Saladin Adversarial Fence - a challenger-based candidate validation mechanism
 5. Furnace / Angel of Death Burn Layer - a hard filter eliminating adversarially-challenged candidates
 6. Three Fates Concordance Arbitration - a three-voter consensus mechanism with temperature diversity
 7. Banyan Metric Value (BMV) 10-Dimension Scoring - a composite quality measurement system
 8. Detective TEAM Four-Gate Root-Cause Analysis - a systematic failure diagnosis mechanism
 9. Andon Cord Re-Plow Loop - Six Sigma stop-the-line applied to cooperative AI substrate growth
 10. Per-Domain Isolation Architecture - fourteen independent domain processing tracks
 11. Starter Chocolate Pre-Seeded Substrate - a cooperative-class onboarding mechanism
 12. MEMORY.md Self-Context Auto-Loading - AI identity injection at every inference call
 13. Six-Folder Substrate Layout - a structured local-first AI knowledge storage architecture
 14. Federation Node Frontier - a cooperative capsule rental and work-and-pay architecture
 15. MemoryWallPyramid Addressing - a recursive soccerball-DAG pocket-universe-pyramid addressing system
 16. Package Store Confectionary Architecture - a tiered cooperative knowledge marketplace
 17. Substitution Rail Triple-Mode Exchange - fiat/marks/barter with structural fiat-conversion prohibition
 18. Cross-Pollination Reciprocity (XP/RP) - at-risk acquisition for high-tier cooperative knowledge packs
 19. Excalibur Active Currents (EAC) Pheromone Decay - substrate priority management function
 20. Catacombs Cold Archive - threshold-based archival with dual-period grace
 21. MMLU-Pro Cooperative-Pipeline Benchmark - reproducible empirical measurement framework
 22. Tab UX Option B with Per-Tab Pin/Hide - single-row Regular/Advanced-toggle interface architecture
 23. Test-Net by Design Ledger Architecture - cooperative-class ledger design with fiat-conversion prohibition
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DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Claim Group 1: Canonical Plow Pipeline Architecture

1.1 A computer-implemented method for growing a cooperative-class local-first artificial intelligence knowledge substrate comprising:

- (a) a Spider stage that locates topic-relevant knowledge units in a local substrate index using topical keyword matching against a domain-scoped eplet store;
- (b) a Sprite stage that retrieves located knowledge units from substrate storage;
- (c) a staggered swarm dispatch stage that queries nine external specialist data sources with a FireGuard time delay of approximately 1.0 seconds between each sequential specialist dispatch;
- (d) a Miner filter stage applying a weight threshold of ≥ 0.6 AND content length threshold of ≥ 100 characters, rejecting any candidate failing either threshold;

- (e) a Saladin adversarial fence stage submitting each Miner-passing candidate to a local AI model with a challenge prompt requesting PASS or CHALLENGE verdict on factual accuracy;
- (f) a Furnace burn stage eliminating all CHALLENGE-verdict candidates from further processing;
- (g) a Three Fates concordance arbitration stage employing three independent voter prompts at temperatures 0.0, 0.2, and 0.4 to synthesize a final answer from Furnace-surviving candidates;
- (h) a Scribe stage recording the result with a ten-dimensional Banyan Metric Value quality score, concordance classification (CONCORDANT, PARTIAL, or DISCORDANT), and outcomes for four quality gates; and
- (i) a Detective TEAM stage performing root-cause analysis on quality gate failures and activating an Andon Cord re-plow loop when any gate fails.

1.2 The method of 1.1 wherein the nine external specialist data sources comprise Wikipedia, Wikidata, StackExchange, arXiv, Wolfram Alpha, OpenAlex, NIST, PubMed, and Common Crawl, each accessed via a dedicated specialist adapter.

1.3 The method of 1.1 wherein the staggered swarm dispatch selects a domain-specific operator subset using a per-domain operator map assigning relevance weights per specialist per academic domain.

1.4 The method of 1.1 wherein Spider-stage knowledge units are injected as additional context into Three Fates voter prompts, enabling arbitration over both external specialist evidence and prior verified substrate knowledge.

1.5 The method of 1.1 wherein the Spider stage uses topical-only query excluding exact-match hits on the current question text, ensuring substrate context enriches domain knowledge without providing direct answer-key lookup.

Claim Group 2: FireGuard Staggered Swarm Dispatch

2.1 A computer-implemented mechanism for rate-limit-aware multi-specialist knowledge acquisition comprising:

- (a) a sequential dispatch loop over an ordered list of specialist data source adapters;
- (b) a configurable FireGuard inter-dispatch delay applied between each pair of sequential specialist dispatches;
- (c) non-blocking parallel collection of specialist responses within each dispatch window; and
- (d) a per-domain operator map selecting specialist subsets based on domain-relevance weights, wherein mathematics/engineering domains receive different subsets than history/law domains.

2.2 The mechanism of 2.1 wherein the FireGuard inter-dispatch delay is approximately 1,000 milliseconds.

2.3 The mechanism of 2.1 wherein specialist adapter failure is non-fatal: the failed adapter's candidates are omitted and processing continues with remaining specialists.

Claim Group 3: Andon Cord Self-Policing Mechanism

3.1 A computer-implemented system for cooperative AI substrate quality enforcement comprising:

- (a) a Detective TEAM component monitoring four quality gates: G1_FACT (any candidates surviving Furnace), G2_CONC (concordance not DISCORDANT), G3_BMV (BMV score meets minimum threshold), and G4_LAT (processing latency within time budget);
- (b) an Andon cord trigger initiating a re-plow loop upon any quality gate failure;
- (c) an operator-widening mechanism expanding the specialist set with additional specialists identified by Detective TEAM as root-cause-relevant;
- (d) a restart-from-question mechanism replaying the full pipeline for the current question with the widened operator set; and
- (e) a retry cap of a configurable maximum (default: three) Andon activations before marking the question PARTIAL.

3.2 The system of 3.1 wherein Detective TEAM applies differentiated widening logic per gate: G1_FACT failure adds wikipedia + commoncrawl; Miner-all-filtered adds arxiv + openalex; Furnace-all-burned adds nist + pubmed; G2_CONC failure adds wikipedia + openalex.

3.3 The system of 3.1 wherein the Andon cord mechanism is modeled after Six Sigma stop-the-line manufacturing discipline applied to cooperative AI substrate growth.

Claim Group 4: BMV 10-Dimension Scoring

4.1 A computer-implemented method for measuring cooperative-class knowledge unit quality comprising a ten-dimensional weighted scoring function evaluating: (a) specialist count score; (b) miner pass-rate; (c) furnace pass-rate; (d) eblet count score; (e) concordance class score (CONCORDANT=100, PARTIAL=60, DISCORDANT=0); (f) latency score; (g) source diversity score; (h) completeness score; (i) quality survival score; and (j) Andon-free score; wherein the composite weighted sum is the Banyan Metric Value (BMV) ranging 0.0-100.0.

4.2 The method of 4.1 wherein CONCORDANT is determined by shared-term ratio ≥ 0.15 across tokenized Three Fates voter answers, PARTIAL by ratio ≥ 0.05 , DISCORDANT otherwise.

Claim Group 5: Per-Domain Isolation Architecture

5.1 A computer-implemented system for parallel cooperative AI knowledge growth comprising fourteen independent domain processing tracks wherein: (a) each domain track maintains no shared mutable state with any other domain track; (b) processing errors, rate limit events, Andon activations, or pipeline failures in one domain do not affect any other domain; (c) each domain track receives an independent specialist operator map; and (d) each domain track writes to a domain-scoped substrate partition.

5.2 The system of 5.1 wherein the fourteen academic domains comprise: Mathematics, Physics, Chemistry, Biology, Engineering, Computer Science, History, Law, Economics, Psychology, Health, Philosophy, Business, and Other.

5.3 The system of 5.1 wherein domain status GREEN is assigned at $\geq$ configurable minimum verified eblets, YELLOW at ≥ 1 eblet below GREEN threshold, RED at 0 eblets.

Claim Group 6: MEMORY.md Self-Context Auto-Loading Architecture

6.1 A computer-implemented method for cooperative AI identity injection comprising:

- (a) generation of a MEMORY.md self-context document at first application launch containing AI instance identity, local substrate layout paths, pocket-universe-pyramid tip address, cooperative architecture affiliation, and Truth-Always discipline declaration;
- (b) automatic loading of the MEMORY.md document as a system prompt prepended to every inference query submitted to the local AI model; and
- (c) file-system watch invalidation of the cached system prompt upon any modification to MEMORY.md, ensuring most-current self-context is used at every inference call without application restart.

6.2 The method of 6.1 wherein the cooperative architecture affiliation declaration specifies: Liana Banyan cooperative architecture identity; Substitution canon prohibiting fiat conversion of cooperative credits; Truth-Always discipline requiring explicit uncertainty acknowledgment; and canonical substrate layout paths enabling the AI instance to locate its own knowledge base.

Claim Group 7: Six-Folder Substrate Layout Architecture

7.1 A computer-implemented data storage architecture for local-first cooperative AI comprising six named directories:

- (a) Substrate - active hot-layer storage for current verified knowledge units;
- (b) Catacomb - cold archive storage for knowledge units whose EAC pheromone decay score has fallen below a configurable minimum threshold;
- (c) Capsules - Federation Node Frontier rental storage for cooperative network participation;
- (d) Vault - user backup storage and IP ledger replica for cooperative canonical hashes;
- (e) MemoryWallPyramid - recursive soccerball-DAG pocket-universe-pyramid addressing storage; and
- (f) TestData - pre-propagated benchmark seed storage.

7.2 The architecture of 7.1 wherein the Catacomb cold archive applies a primary threshold period and a configurable

grace period (default 90 + 30 days) before permanent archival.

Claim Group 8: Starter Chocolate Pre-Seeded Substrate

8.1 A computer-implemented method for cooperative AI onboarding comprising:

- (a) packaging a pre-verified set of cooperative-class knowledge units designated Starter Chocolate as an extra resource within the application installer;
- (b) automatically extracting and installing Starter Chocolate to the active Substrate directory upon first application launch, detected via absence of substrate directory or idempotency marker; and
- (c) preventing duplicate installation via an idempotency marker file written upon successful first installation.

8.2 The method of 8.1 wherein each Starter Chocolate knowledge unit is tagged with source identifier "starter_chocolate" for provenance tracking, distinguishing it from dynamically-grown substrate knowledge units.

Claim Group 9: Federation Node Frontier Architecture

9.1 A computer-implemented system for cooperative AI network participation comprising:

- (a) a Capsule rental mechanism enabling cooperative node owners to rent substrate processing capacity to cooperative participants, denominated in cooperative Marks credits;
- (b) a Thorax-encrypted transport layer securing inter-node communication;
- (c) a work-and-pay mechanism wherein participants earn Marks for knowledge units contributed and expend Marks to access capsule rental capacity; and
- (d) a Medallion provenance stamp on each knowledge unit identifying its originating cooperative node.

Claim Group 10: MemoryWallPyramid Addressing

10.1 A computer-implemented addressing system for distributed cooperative AI knowledge comprising:

- (a) a pocket-universe-pyramid structure organized as a recursive soccerball-directed acyclic graph with branching factor six;
- (b) a nine-hexadecimal-digit tip address uniquely identifying each user's pyramid instance;
- (c) a 6^N reach property enabling navigation to any knowledge unit within N hops from the pyramid tip; and
- (d) lazy resolution of pyramid nodes upon first access.

Claim Group 11: Package Store Confectionary Architecture

11.1 A computer-implemented cooperative knowledge marketplace comprising:

- (a) a Bakery stall offering full-corpus knowledge packs (Breads);
- (b) a Confectionary stall offering curated domain-specific knowledge packs (Chocolates);
- (c) a Kitchen stall offering custom knowledge preparation services; and
- (d) a cross-pollination reciprocity mechanism requiring exchange points at risk for high-tier pack acquisition.

11.2 The marketplace of 11.1 wherein each knowledge pack is stamped with an immutable Medallion provenance identifier linking to the originating cooperative node and IP ledger.

Claim Group 12: Substitution Rail Triple-Mode Exchange

12.1 A computer-implemented cooperative exchange system comprising three substitution modes:

- (a) Fiat mode for conventional currency transactions at Cost+20% rates;
 - (b) Marks mode for cooperative credit transactions without fiat conversion, wherein Marks circulate exclusively within the cooperative; and
 - (c) Barter mode for direct goods-and-services exchange between cooperative members;
- wherein cooperative credits (Marks, Joules, Credits) are prohibited from fiat conversion by structural cooperative bylaw

enforced at the transaction validation layer.

Claim Group 13: GPQA Diamond Cooperative-Pipeline Benchmark

13.1 A computer-implemented benchmarking method comprising:

- (a) a Bare Diamond mode executing zero-shot inference without substrate enrichment on the GPQA Diamond 198-question graduate-level reasoning benchmark;
- (b) a Cooperative-Pipeline Diamond mode executing the canonical plow pipeline with substrate enrichment on the same benchmark; and
- (c) a cooperative-architecture lift metric computed as the percentage-point difference between Cooperative-Pipeline and Bare Diamond accuracy.

13.2 The method of 13.1 wherein the MMLU-Pro 70-question canonical benchmark is additionally used for empirical receipt generation, with 97.1% (68/70) on consumer hardware with no paid API keys constituting a reduction-to-practice receipt.

Claim Group 14: Excalibur Active Currents Pheromone Decay

14.1 A computer-implemented method for substrate knowledge priority management comprising:

- (a) an EAC pheromone decay function assigning a recency score to each substrate knowledge unit based on elapsed time since last access or verification;
- (b) periodic recalculation of decay scores across the active Substrate partition with units below a configurable minimum threshold marked for Catacomb migration;
- (c) automatic elevation of a knowledge unit decay score upon each access, query hit, or successful concordance arbitration use; and
- (d) a configurable primary threshold period and grace period before permanent archival (default: 90 days primary + 30 days grace).

Claim Group 15: MMLU-Pro Disk-Backed Benchmark Architecture

15.1 A computer-implemented benchmark execution architecture comprising:

- (a) a sealed question bank from TIGER-Lab MMLU-Pro stored as disk-backed JSON partitioned by fourteen academic domains;
 - (b) a deterministic stride-sampling algorithm ensuring reproducible question selection across benchmark runs;
 - (c) per-domain result isolation enabling partial reruns without affecting other domains; and
 - (d) a SHA-256-hashed receipt file written to disk upon completion containing per-domain accuracy, aggregate accuracy, substrate growth count, and machine identification.
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CLAIMS

The detailed description above presents the claims framework for this provisional patent application. Formal independent and dependent claims will be crystallized in the corresponding non-provisional application. The applicant reserves the right to file a non-provisional application claiming priority to this provisional application for all innovations disclosed herein.

BRIEF DESCRIPTION OF DRAWINGS

Drawings to be submitted with non-provisional application:

- Figure 1: Canonical Plow Pipeline Architecture (nine-stage sequential flow)

- Figure 2: Andon Cord Re-Plow Loop (state machine with three-retry cap)
 - Figure 3: Per-Domain Isolation Architecture (fourteen parallel tracks)
 - Figure 4: Six-Folder Substrate Layout (directory hierarchy)
 - Figure 5: MemoryWallPyramid Addressing (soccerball-DAG recursive structure)
 - Figure 6: Package Store Confectionary (three-stall marketplace)
 - Figure 7: Substitution Rail Triple-Mode Exchange
 - Figure 8: MEMORY.md Self-Context Auto-Loading cycle
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ABSTRACT

A cooperative-architecture AI system and method for local-first knowledge substrate growth on consumer hardware. The system employs a nine-stage canonical plow pipeline: Spider, Sprite, nine-specialist FireGuard-staggered swarm with per-domain operator mapping, dual-threshold Miner filter, Saladin adversarial fence, Furnace burn, Three Fates three-voter concordance arbitration, Scribe BMV scoring with four-gate quality evaluation, and Detective TEAM with Andon cord stop-the-line re-plow loop. Per-domain isolation across fourteen academic domains prevents cross-contamination. Additional innovations: MEMORY.md self-context auto-loading with file-system watch; six-folder substrate layout (Substrate/Catacomb/Capsules/Vault/MemoryWallPyramid/TestData); Starter Chocolate pre-seeded cooperative onboarding; Federation Node Frontier capsule rental with Thorax-encrypted inter-node transport; MemoryWallPyramid recursive soccerball-DAG pocket-universe addressing; Package Store Confectionary three-stall cooperative knowledge marketplace; Substitution Rail triple-mode exchange (Fiat/Marks/Barter) with structural fiat-conversion prohibition; Excalibur Active Currents pheromone decay; and MMLU-Pro disk-backed cooperative benchmark architecture. Empirical receipt: 97.1% (68/70 MMLU-Pro) on consumer hardware, no paid API keys, 316 verified eblets, 14/14 domain GREEN status.

Liana Banyan Corporation - Inventor: J. Jones - Provisional Patent Application

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15 Claim Groups - 23 Innovation Areas - Incremental over Provisional Applications 1-21